# Reviewer Pack — Minimal Rank of Free Probability Entanglement Dimensions vs …

Entry 77614a2866d2 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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## Minimal Rank of Free Probability Entanglement Dimensions vs Communication Complexity for Disjointness

**Entry ID**: `77614a2866d2` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-22 23:05:48 UTC

### 1. Conjecture

**Field A** (mathematical branch): Free Probability Theory **Field B** (complexity object): Communication Complexity: Disjointness

**Statement**:

> ['For any instance of the disjointness problem with n inputs, the minimal rank of a free probability representation of the associated bipartite state is lower bounded by Ω(n).', 'Equivalently, for all such instances, there exists a bipartite state with entanglement dimension at least Ω(n) that can represent the partial function defining disjointness.', 'This lower bound holds with respect to the randomized communication complexity of the disjointness problem.']

**Rationale (proposer's reasoning)**:

> ['Free probability theory offers tools for studying quantum information, and its representations have been linked to entanglement in bipartite states. If such an invariant can be used to characterize classical communication complexity, it could provide a new avenue for understanding the limits of classical computing.', 'The use of free probability might reveal non-trivial structures that are not apparent from standard classical complexity theory approaches, potentially leading to improved lower bounds.', 'This conjecture bridges the gap between quantum information and classical computation by applying quantum-inspired mathematics to a classic problem in communication complexity.']

**Taxonomy category**: `COMM_DISJ` (status at proposal time: )

### 2. Pre-registration (Popper-style)

**Hash (SHA-256 prefix)**: `f5199efd2ff8c08d`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

**Acceptance criterion**:

> The conjecture is supported if, for 30 independent random seeds, the minimum entanglement dimension of free probability representations across all instances meets or exceeds

Ω(n), where n ≤ 40, and the average entanglement dimension is within ±10% of the lower bound.

## 3. Barrier filter (F1)

**Final**: PASS

| Barrier | Verdict | Confidence | LLM₁ | LLM₂ |
|---|---|---|---|---|
| RELATIVIZATION | UNCERTAIN | 1.00 | HITS | UNCERTAIN |
| NATURAL_PROOFS | SAFE | 0.50 | UNCERTAIN | UNCERTAIN |
| ALGEBRIZATION | SAFE | 0.50 | UNCERTAIN | UNCERTAIN |
| KARP_LIPTON | SAFE | 1.00 | SAFE | UNCERTAIN |

## 4. Novelty audit

**Verdict**: ADJACENT_OK against 3 hits across arXiv + Semantic Scholar + ECCC

**Search queries** (3): - "free probability theory" AND "communication complexity" AND "disjointness" - "minimal rank" free probability AND entanglement dimension disjointness - "randomized communication complexity" AND lower bound entanglement dimension free probability"

**Top relevant hits considered**: - [http://arxiv.org/abs/1201.1666v1] A direct product theorem for bounded-round public-coin randomized communication complexity - [http://arxiv.org/abs/2111.10436v2] A counter-example to the probabilistic universal graph conjecture via randomized communication complexity - [http://arxiv.org/abs/2104.11275v1] The Randomized Communication Complexity of Randomized Auctions

## 5. Empirical test harness

**Multi-seed protocol**: 5 seeds (11, 23, 37, 53, 71)
**Sandbox**: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

### 5.1 Generated Python source

```python
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_disjointness_instance(n):
        inputs = [random.randint(0, 1) for _ in range(n)]
        return inputs

    def calculate_entanglement_dimension(instance):
        n = len(instance)
        if n == 1:
            return 1
        elif n == 2:
            return 2
        else:
            # Simplified example: entanglement dimension is n
            return n
```

```
    instances_tested = 0
    min_entanglement_dimension = float('inf')

    for _ in range(30):
        instance = generate_disjointness_instance(random.randint(5, 40))
        entanglement_dimension = calculate_entanglement_dimension(instance)
        if entanglement_dimension < min_entanglement_dimension:
            min_entanglement_dimension = entanglement_dimension
        instances_tested += 1

    conjecture_holds = min_entanglement_dimension >= random.randint(5, 40) / 2
    counterexample = "" if conjecture_holds else f"n={min_entanglement_dimension}, entanglement_di

    return {
        "metric_name": "Minimal Entanglement Dimension",
        "metric_value": min_entanglement_dimension,
        "instances_tested": instances_tested,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    seeds = list(map(int, sys.argv[1:])) if len(sys.argv) > 1 else [2, 3, 5, 7, 11, 13, 17, 19, 23

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    std_metric_value = math.sqrt(sum((r["metric_value"] - mean_metric_value) ** 2 for r in results
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{r['counterexample']}\" first_failing_seed={fir
    else:
        print("RESULT: INCONCLUSIVE")
```

## 6. Per-seed results

*(no seed-level data — test crashed before producing TRIAL: lines)*

## 7. Test stdout (last 2KB)

```
conjecture_holds': False, 'counterexample': 'n=5, entanglement_dimension=5'}
TRIAL: {'metric_name': 'Minimal Entanglement Dimension', 'metric_value': 6, 'instances_tested': 30, '
TRIAL: {'metric_name': 'Minimal Entanglement Dimension', 'metric_value': 13, 'instances_tested': 30,
TRIAL: {'metric_name': 'Minimal Entanglement Dimension', 'metric_value': 5, 'instances_tested': 30, '
TRIAL: {'metric_name': 'Minimal Entanglement Dimension', 'metric_value': 6, 'instances_tested': 30, '
TRIAL: {'metric_name': 'Minimal Entanglement Dimension', 'metric_value': 6, 'instances_tested': 30, '
TRIAL: {'metric_name': 'Minimal Entanglement Dimension', 'metric_value': 5, 'instances_tested': 30, '
TRIAL: {'metric_name': 'Minimal Entanglement Dimension', 'metric_value': 5, 'instances_tested': 30, '
```

```
TRIAL: {'metric_name': 'Minimal Entanglement Dimension', 'metric_value': 5, 'instances_tested': 30, '

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c25bcaf9.py", line 72, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c25bcaf9.py", line 72, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                              ~^^^^^^^^
KeyError: 'seed'
```

## 8. Critic adversarial review

**Critic verdict**: CHALLENGE

**Critic reasoning**:

> skipped: test crashed before producing data

## 9. Final verdict & safety rail

**Verdict**: INCONCLUSIVE

**Reasoning**:

> The test code crashed before producing data, which prevents a definitive evaluation of the conjecture. | next: Investigate and fix the crash in the test code to proceed with the verification of the conjecture.

## 11. Audit log (LLM calls)

**Total LLM calls**: 10

| # | Phase | Provider | Model | Tokens in | out | Latency (ms) |
|---|---|---|---|---|---|---|
| 1 | propose | ollama_remote | glm4:latest | 0 | 0 | 14986 |
| 2 | propose | ollama_remote | glm4:latest | 0 | 0 | 16841 |
| 3 | preregistration | ollama_remote | glm4:latest | 0 | 0 | 9661 |
| 4 | novelty | ollama_remote | glm4:latest | 0 | 0 | 9444 |
| 5 | novelty | ollama_remote | glm4:latest | 0 | 0 | 9895 |
| 6 | test_gen | ollama_remote | qwen2.5-coder:7b | 0 | 0 | 13105 |
| 7 | test_gen | ollama_remote | qwen2.5-coder:7b | 0 | 0 | 9564 |
| 8 | test_gen | ollama_remote | qwen2.5-coder:7b | 0 | 0 | 8430 |
| 9 | test_gen | ollama_remote | qwen2.5-coder:7b | 0 | 0 | 7797 |
| 10 | judge | ollama_remote | glm4:latest | 0 | 0 | 11559 |

**Totals**: 0 input tokens, 0 output tokens, ~$0.0000 cost, 111283 ms total latency. Provider mix: {'ollama_remote': 10}

*(full prompt+response transcripts available in `research/audit/77614a2866d2.jsonl`)*

## 12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice

3. The full LLM transcripts are in `research/audit/77614a2866d2.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

**Sandbox file**: `research/pvsnp_sandbox/test_*.py` (per-cycle)
**Replay tarball**: `research/replay/77614a2866d2.tar.gz` (if generated)